

```

%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;
scuts_path = 'e:\SHARE\Fe\Data\analysis\07_fit_with_multicut_allJ\sel_cuts';
Cutfitfile = fullfile(scuts_path, 'GP_FFCutsFits.mat');

if ~exist('GP_FFCutsFits', 'var')
    ld = load(Cutfitfile);
    GP_FFCutsFits = ld.fit_src_struct;
end

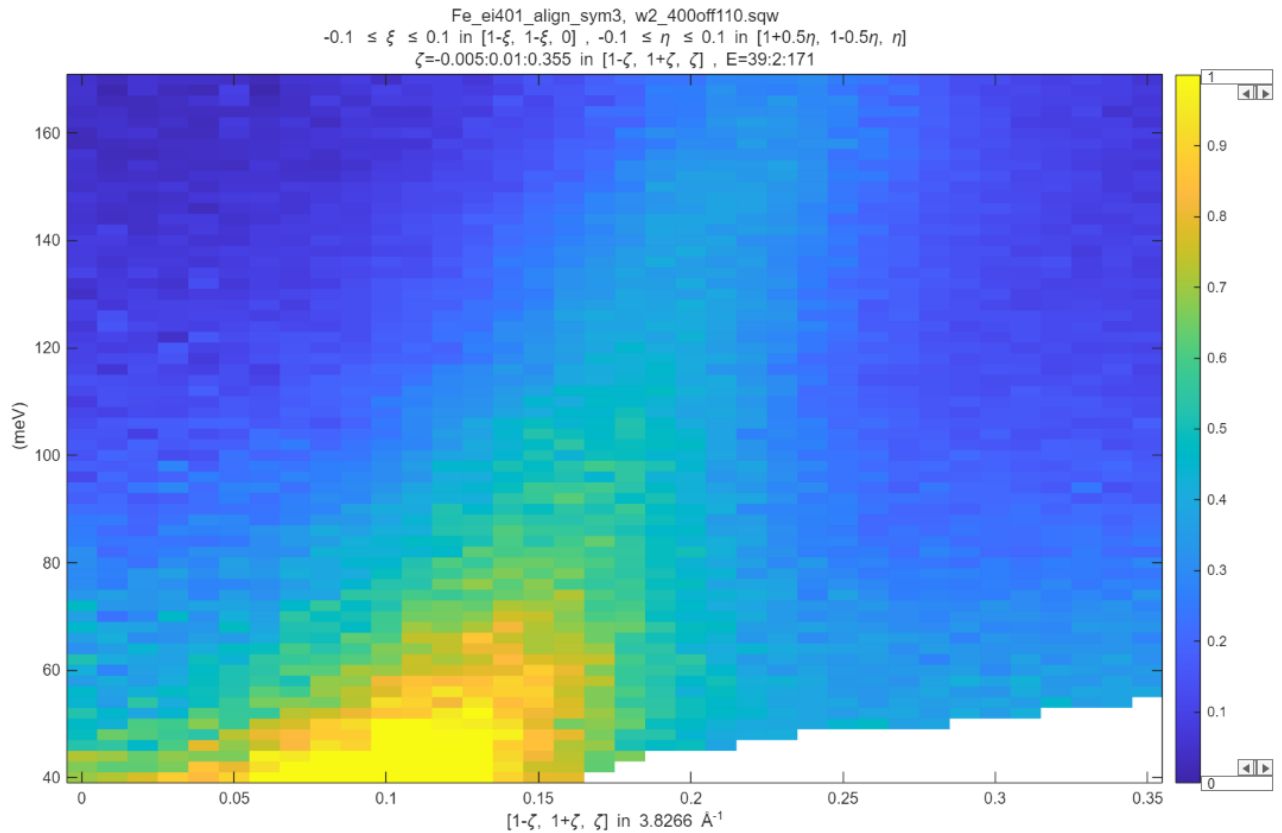
```

```

selectCuts= 'Ei400_off110GPminFF';
cuts_list = GP_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end

%mi = maps_instrument(200,600,'S');
mi = maps_instrument(401,600,'S');
%mi = maps_instrument(786,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0], 'cuboid', [0.04,0.03,0.02]);
for i=1:numel(cuts_list)
    cuts_list{i} = cuts_list{i}.set_instrument(mi);
    cuts_list{i} = cuts_list{i}.set_sample(sample);
    plot(cuts_list{i}); lz 0 1; keep_figure;
end

```



```

dir_name = "GP";

dE_step = 2; %original energy transfer step data were binned to. No point in going
finer
half_dE = 5; % half width of data binning

en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(30,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 110;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
GP_FFCutsFits.(selectCuts) = cuts_list;

```

```

GP_FFCutsFits = fit_multicutsFF_along_direction(...
    GP_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);

```

```

*****
**** En = 44±5 going to 164
*****

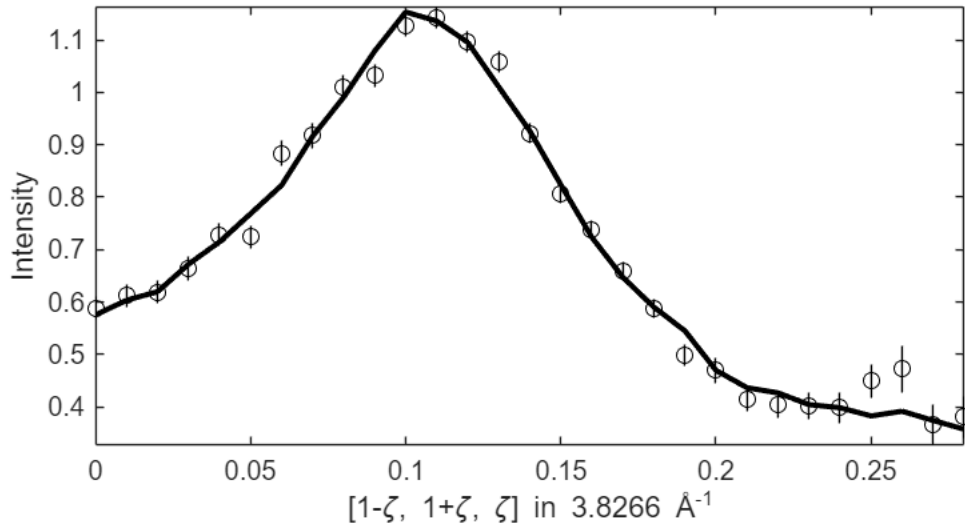
```

Fe_ei401_align_sym3, w2_400off110.sqw

S=0.29±0.0076; J0= 6.7±0.033; gamma=4.41293±0.0909923

.1 ≤ ξ ≤ 0.1 in [1-ξ, 1-ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1+0.5η, 1-0.5η, η] , 39 ≤ E

ζ=-0.005:0.01:0.355 in [1-ζ, 1+ζ, ζ]



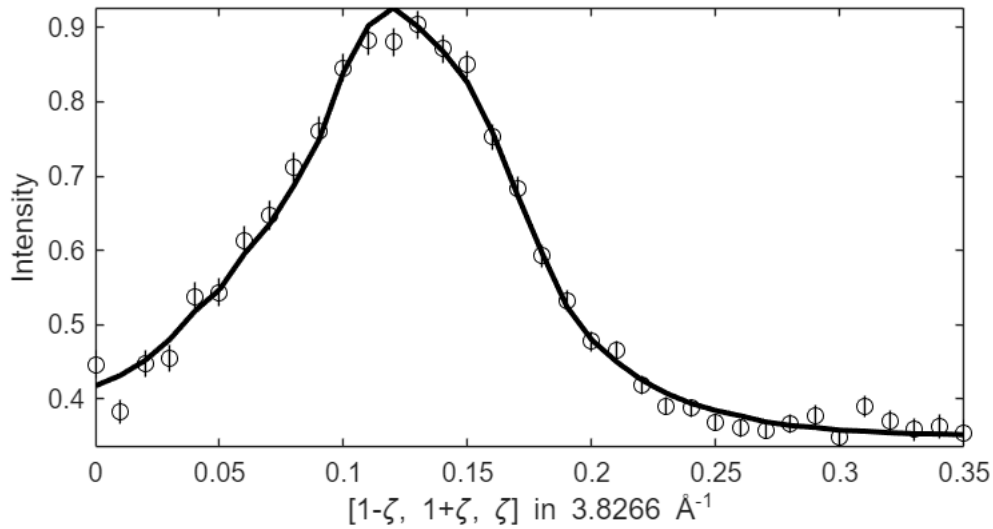
**** En = 54±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw

S=0.19±0.0063; J0= 6±0.029; gamma=3.04936±0.631903

.1 ≤ ξ ≤ 0.1 in [1-ξ, 1-ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1+0.5η, 1-0.5η, η] , 49 ≤ E

ζ=-0.005:0.01:0.355 in [1-ζ, 1+ζ, ζ]



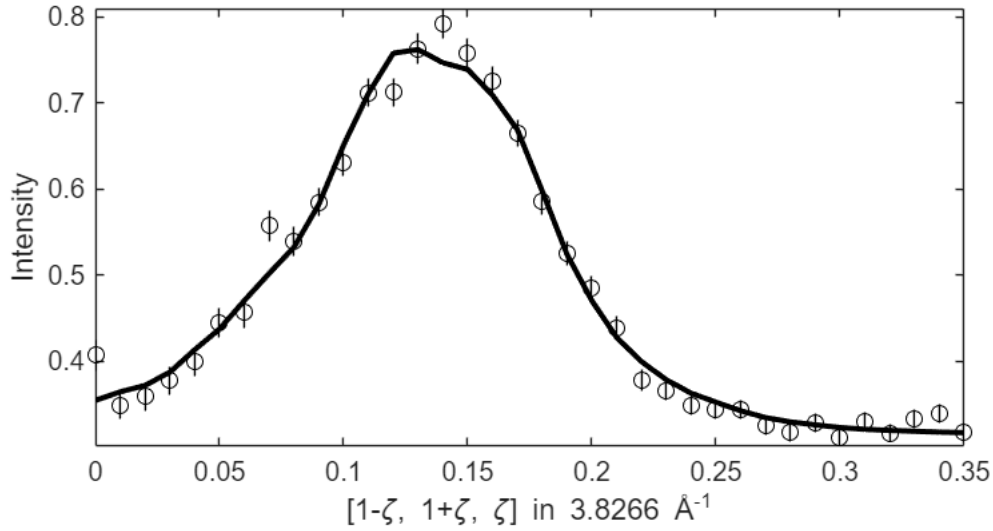
**** En = 64±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw

S=0.16±0.0032; J0= 5.7±0.033; gamma=3.88113±0.0603236

.1 ≤ ξ ≤ 0.1 in [1-ξ, 1-ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1+0.5η, 1-0.5η, η] , 59 ≤ E

ζ=-0.005:0.01:0.355 in [1-ζ, 1+ζ, ζ]



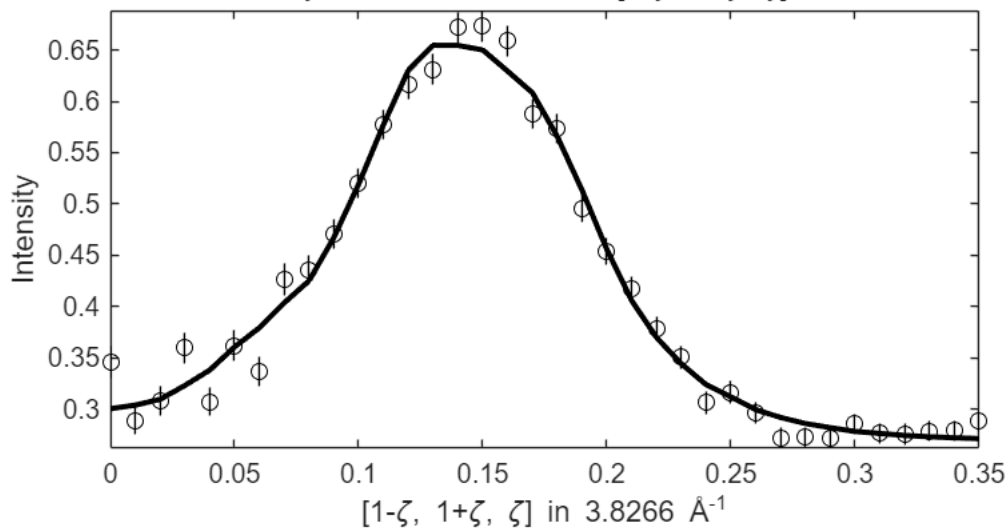
**** En = 74±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw

S=0.15±0.0033; J0= 5.4±0.031; gamma=5.00896±0.0840584

.1 ≤ ξ ≤ 0.1 in [1-ξ, 1-ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1+0.5η, 1-0.5η, η] , 69 ≤ E

ζ=-0.005:0.01:0.355 in [1-ζ, 1+ζ, ζ]



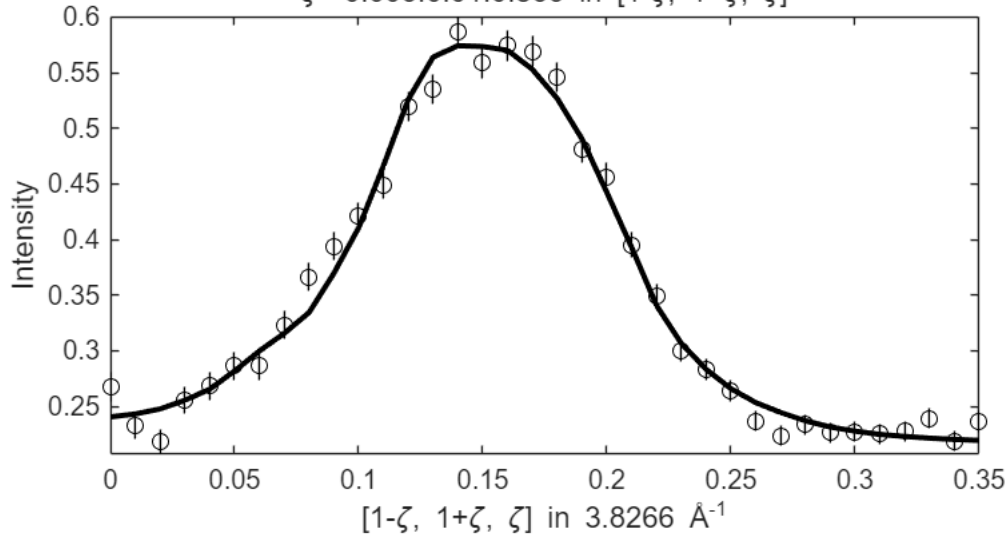
**** En = 84±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw

S=0.15±0.0034; J0= 5.1±0.025; gamma=6.00991±0.105928

.1 ≤ ξ ≤ 0.1 in [1-ξ, 1-ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1+0.5η, 1-0.5η, η] , 79 ≤ E

ζ=-0.005:0.01:0.355 in [1-ζ, 1+ζ, ζ]



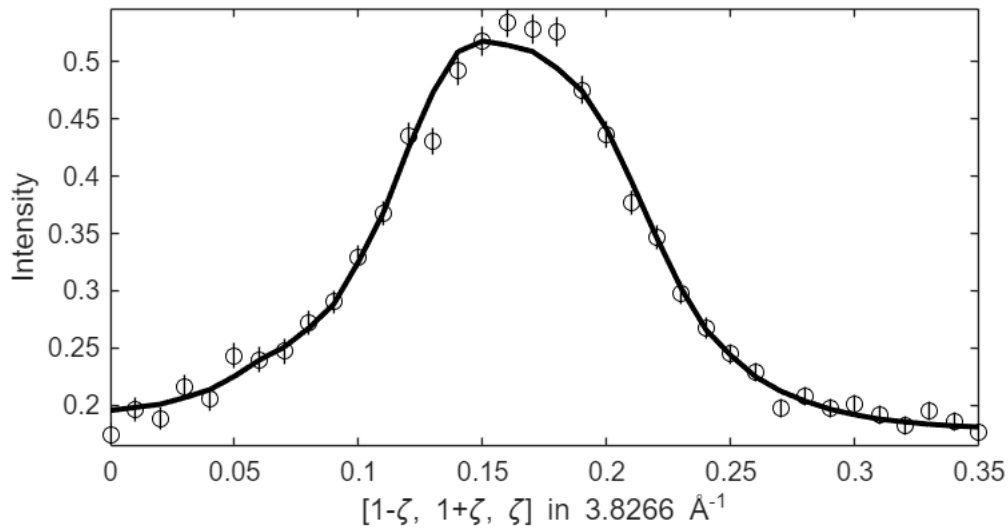
**** En = 94±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw

S=0.14±0.0026; J0= 4.9±0.02; gamma=7.35711±0.0975416

.1 ≤ ξ ≤ 0.1 in [1-ξ, 1-ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1+0.5η, 1-0.5η, η] , 89 ≤ E

ζ=-0.005:0.01:0.355 in [1-ζ, 1+ζ, ζ]



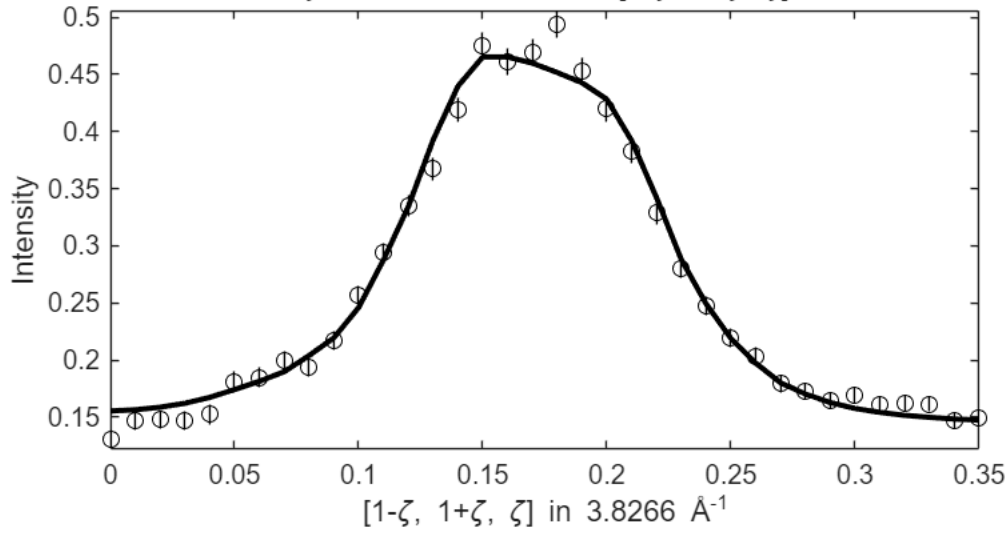
**** En = 104±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw

S=0.13±0.0025; J0= 4.7±0.019; gamma=6.61216±0.104068

$1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $99 \leq E$:

$\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



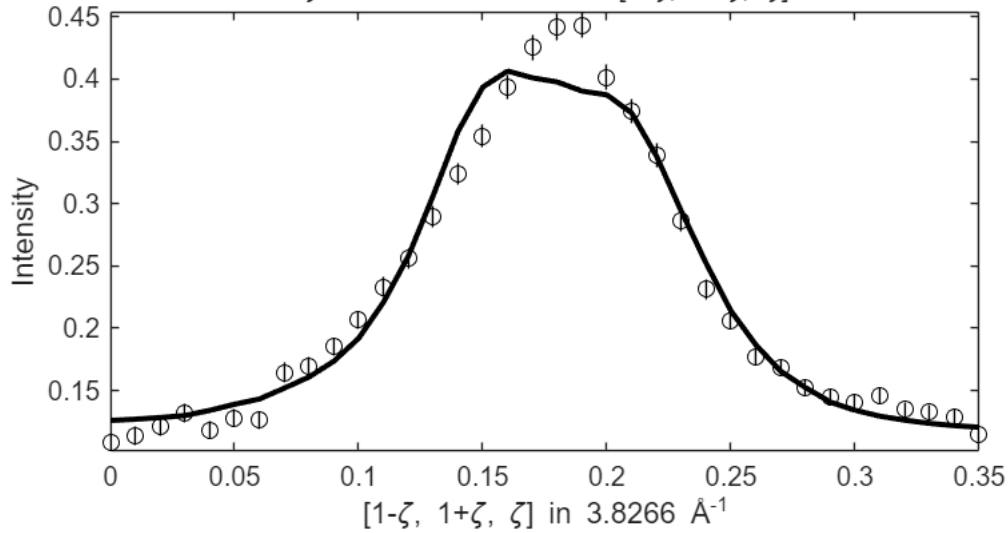
**** En = 114±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw

S=0.13±0.0077; J0= 4.5±0.028; gamma=8.56076±1.84297

$1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $109 \leq E$

$\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



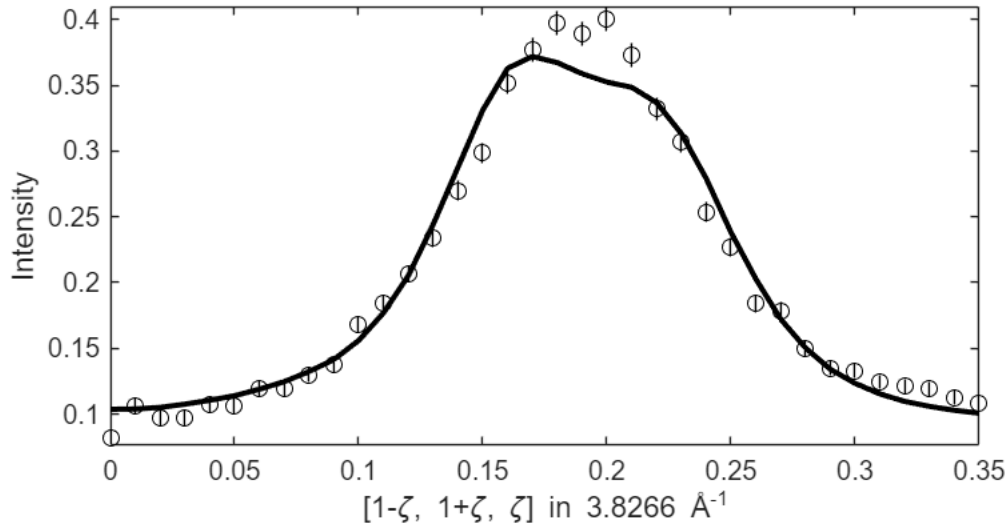
**** En = 124±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw

S=0.14±0.004; J0= 4.3±0.024; gamma=12.0339±0.33578

$1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $119 \leq E$

$\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



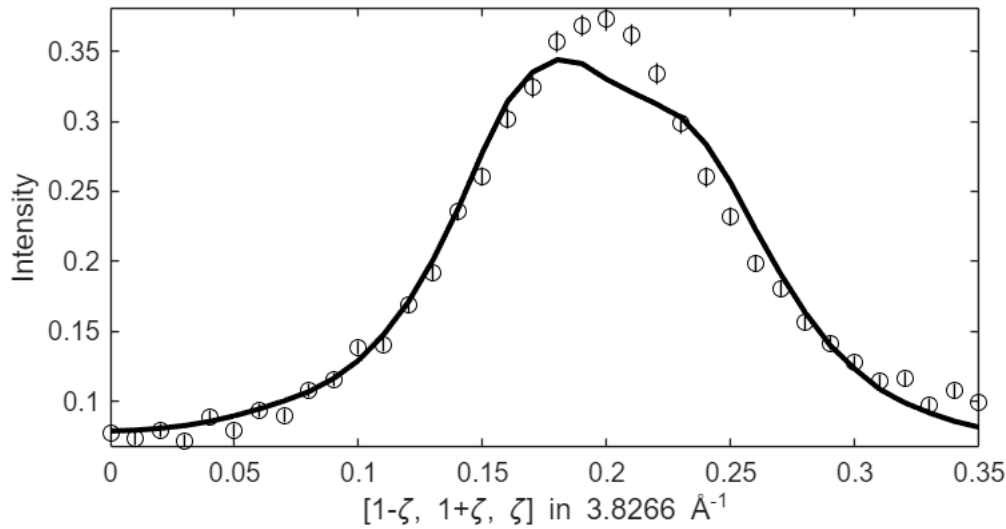
**** En = 134±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw

S=0.17±0.008; J0= 4.1±0.024; gamma=18.8874±1.27967

$1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $129 \leq E$

$\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



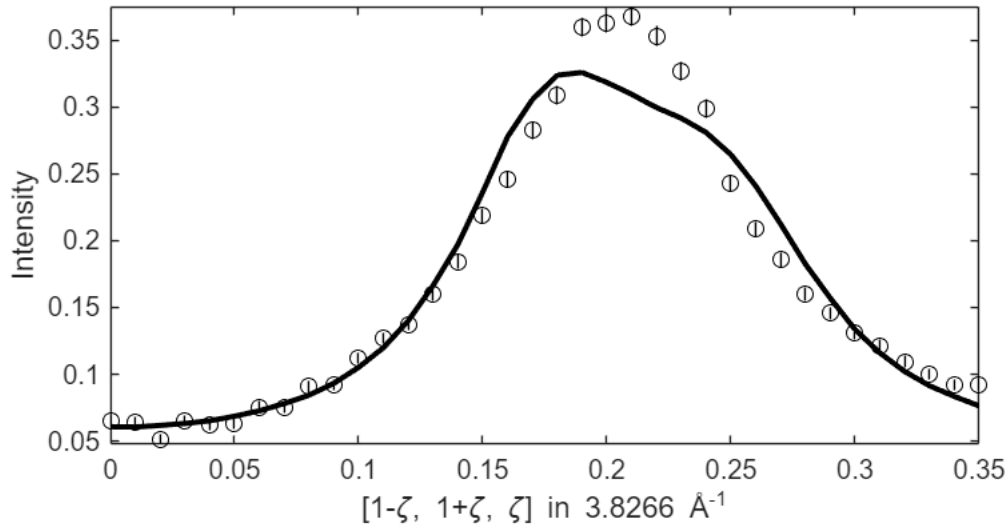
**** En = 144±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw

S=0.19±0.015; J0= 4±0.03; gamma=23.1285±3.38831

$1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $139 \leq E$

$\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



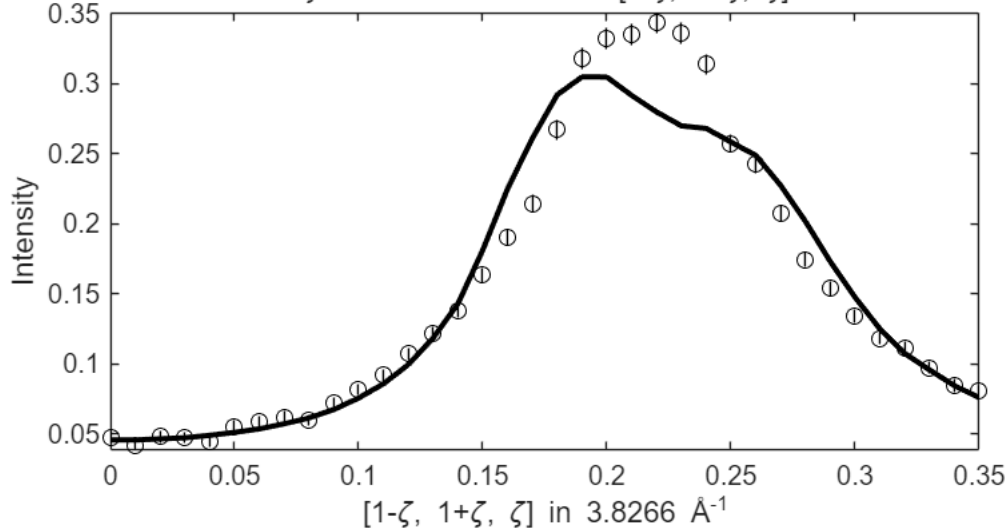
**** En = 154±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw

S=0.19±0.012; J0= 3.8±0.033; gamma=22.8581±3.12219

$1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $149 \leq E$

$\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



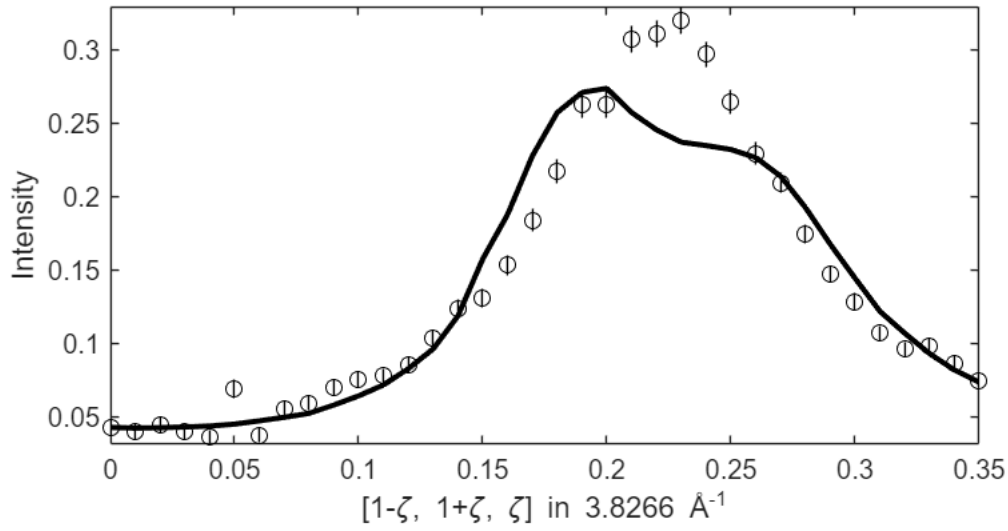
**** En = 164±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw

$S=0.17\pm0.014$; $J0= 3.7\pm0.035$; $\gamma=22.8214\pm3.69606$

$1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $159 \leq E$

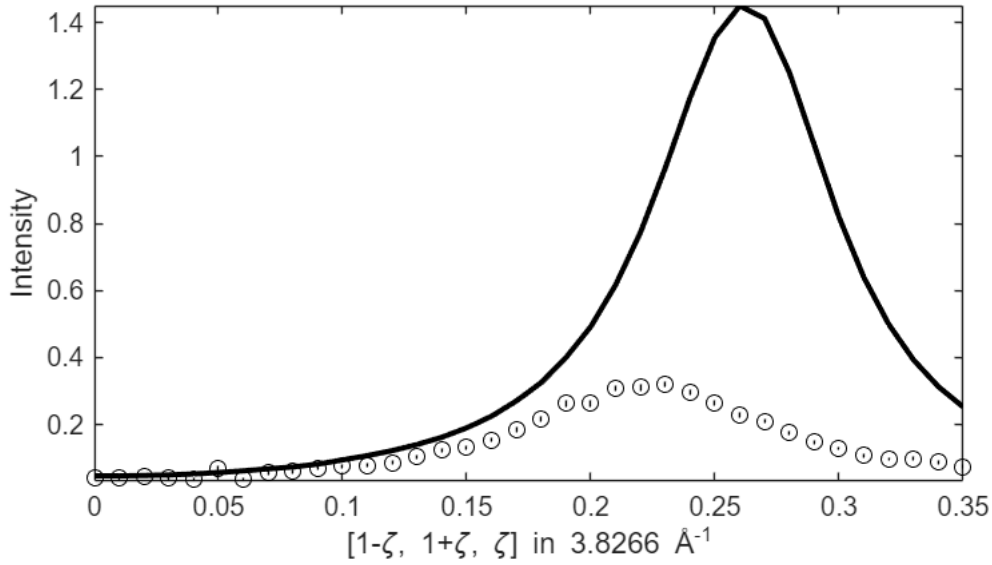
$\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

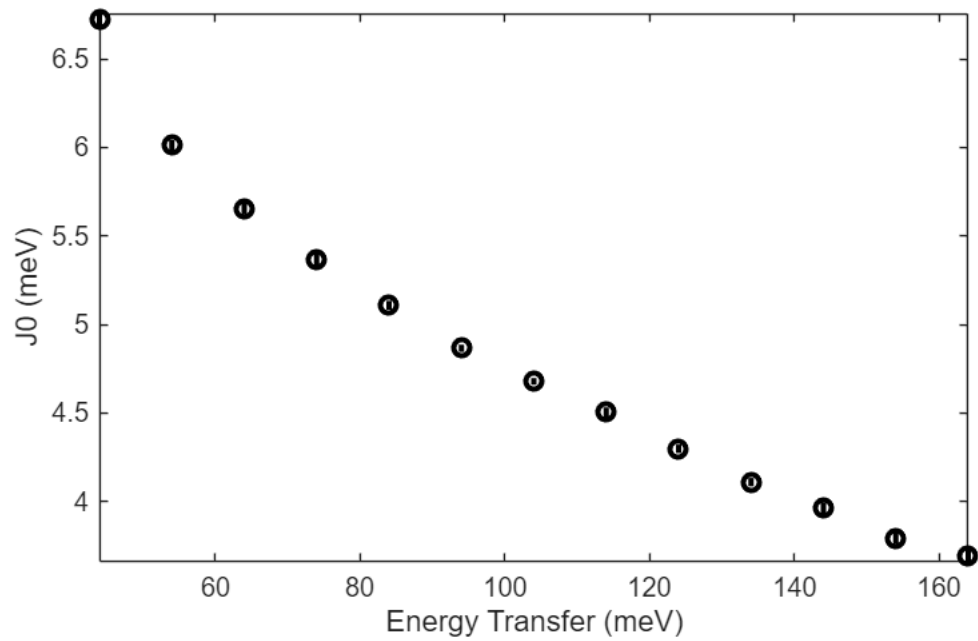
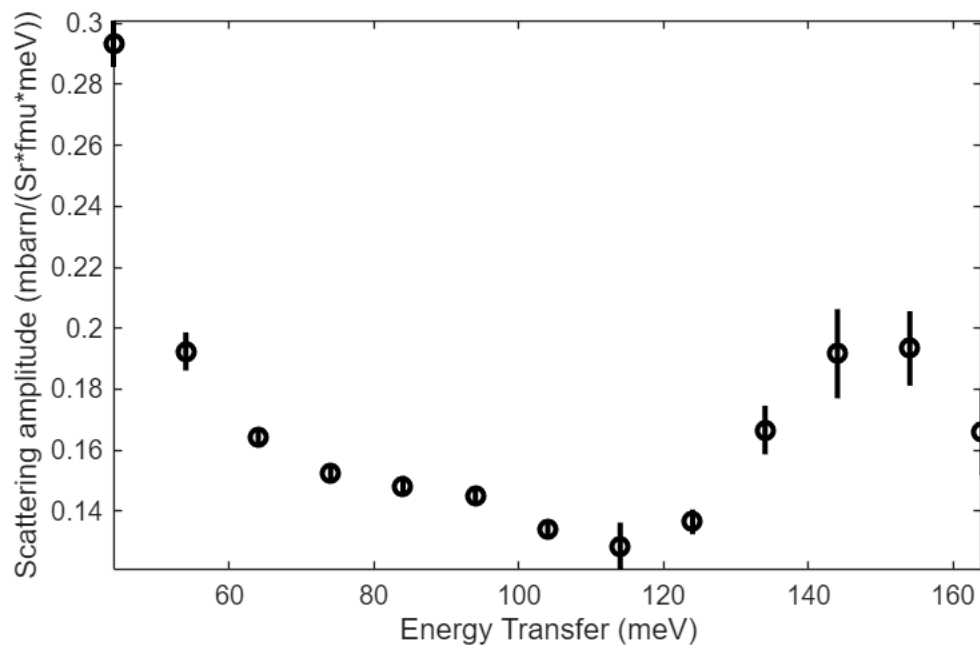


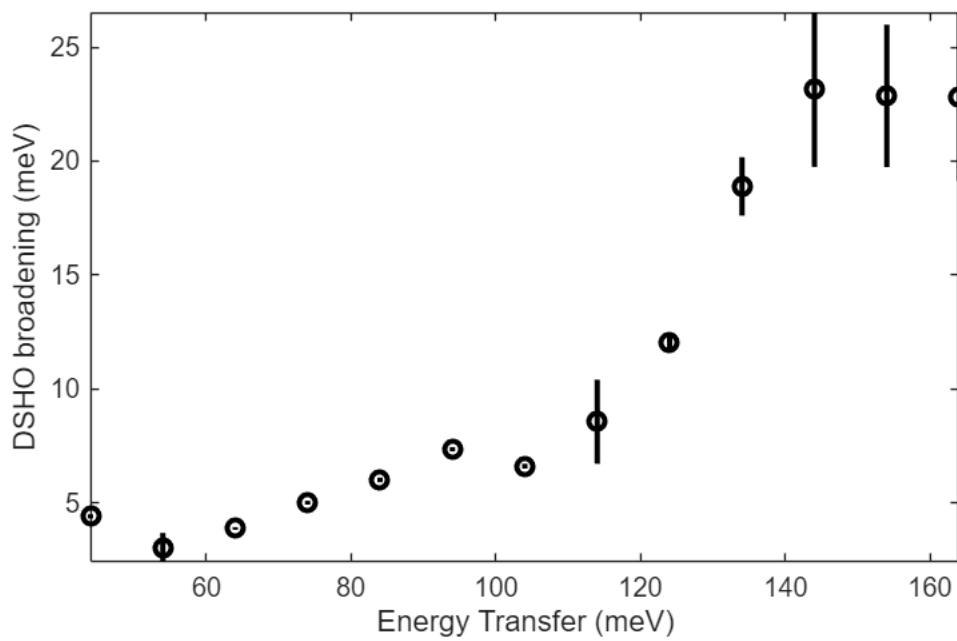
Fe_ei401_align_sym3, w2_400off110.sqw

$1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $159 \leq E$

$\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



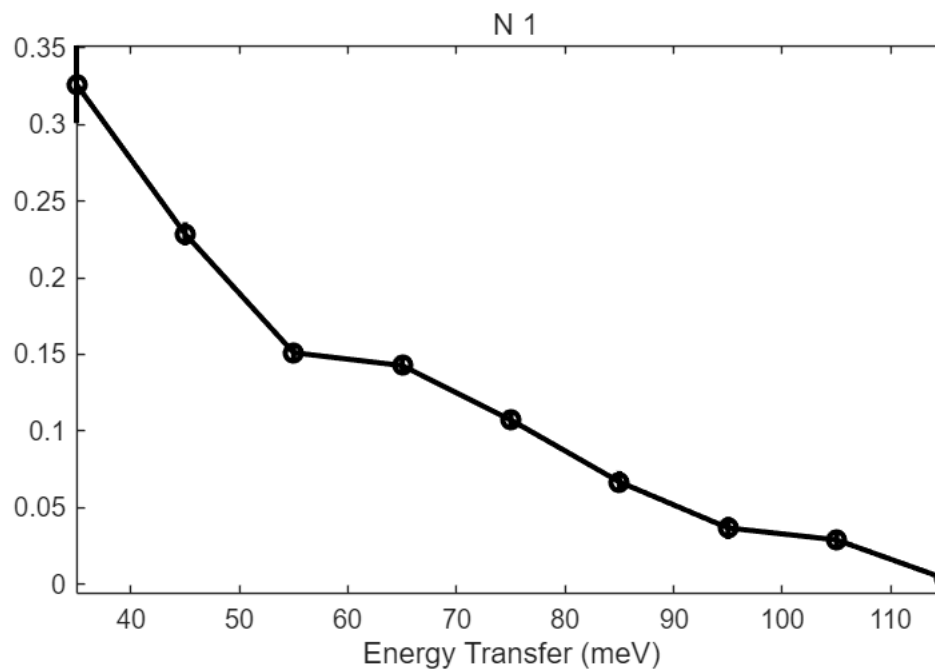




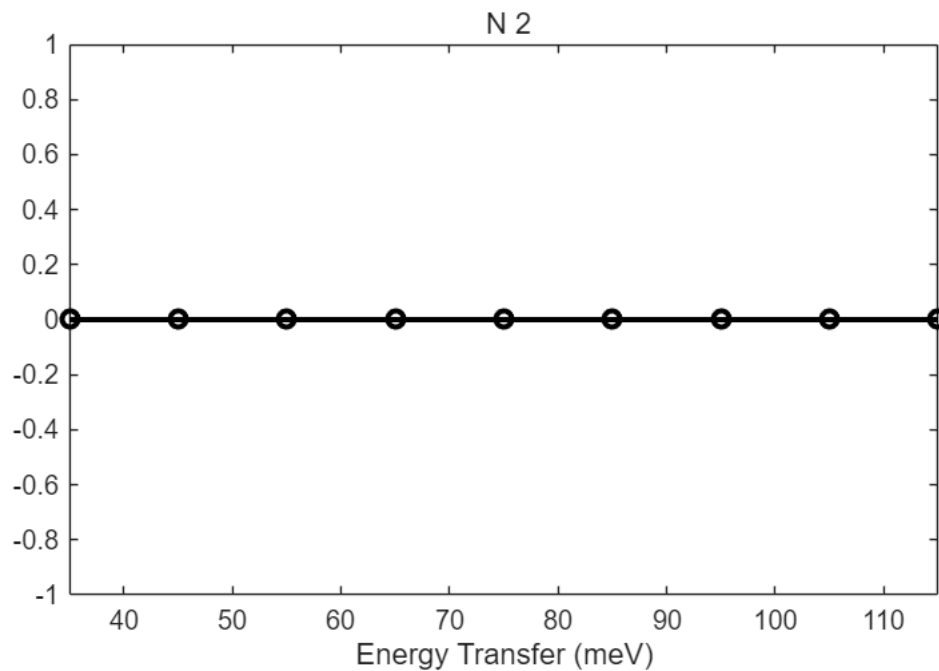
```
c1 = GP_FFCutsFits.(selectCuts);
GP_FFCutsFits.(selectCuts) = [c1{:}];
```

```
beep
beep
beep
```

```
all_fit_par = GP_FFCutsFits.Ei200_off110GPmiff_all_fit_par;
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;
```



```
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_GP.mlx';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_GP_',selectCuts,'.pdf']);
export(thisFile,targ_file);
```

The value of 'Destination' is invalid. It must satisfy the function: maybeFileOrPath.

```
fit_src_struct = GP_FFCutsFits;
save(Cutfitfile,"fit_src_struct",-v7.3')
```